

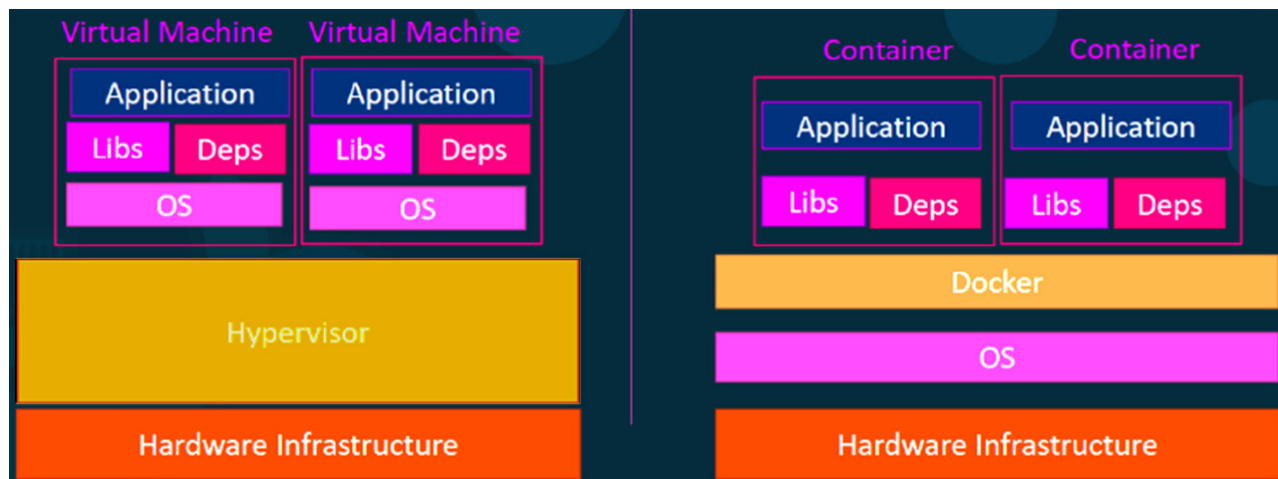


Figure 1: VM vs Docker container

Why Docker is popular

Docker provides a lightweight virtualization solution with zero overhead. It is currently widely used across a variety of sectors. Containers make it possible for developers to deploy a program along with all of its requirements and configurations, including libraries and other dependencies, as a single package, which can then be distributed to end users. Docker is gaining popularity in the world of technology for a variety of reasons, a few of which are mentioned below:

- Efficient resource usage
- Simple configuration
- Enhanced developer productivity
- Provides application isolation
- Fast application development

Installing Docker on Ubuntu/Linux

Before installing the Docker Engine on a new host system, we must configure the Docker repository. The user can then update and install Docker from the repository. The steps for installing Docker and running the example application are listed below.

Step 1: Perform an update on the package index for *apt*, and then install the necessary modules to enable *apt* to use HTTPS archives.

```
$ apt-get -y install apt-transport-https ca-certificates curl
```

Step 2: Download GPG key for Docker.

```
$ curl -fsSL https://download.docker.com/linux/ubuntu/gpg |  
sudo apt-key add -
```

Step 3: Configure the repository with the following command:

```
$ add-apt-repository "deb [arch=amd64] https://download.  
docker.com/linux/ubuntu $(lsb_release -cs) stable"
```

Step 4: Once you have downloaded your *apt* package, install Docker Engine by executing the following commands:

```
$ apt-get update  
$ apt-get -y install docker-ce
```

Step 5: Now verify the installation. Executing the Ubuntu image will allow you to confirm that the Docker Engine was successfully installed.

```
$ Docker pull ubuntu
```

Docker pulls a command to download the required image from the Docker hub, as shown in Figure 2.

Step 6: NGINX is free software used for many different purposes, including web serving, media streaming, reverse proxying, caching, and load balancing. Use the command given below to download the NGINX Docker container image:

```
$ Docker container run Nginx
```

```
root@k-master:/home/osboxes# docker pull ubuntu  
Using default tag: latest  
latest: Pulling from library/ubuntu  
d51af753c3d3: Pull complete  
fc878cd0a91c: Pull complete  
6154df8ff988: Pull complete  
fee5db0ff82f: Pull complete  
Digest:  
sha256:747d2dbbaae995098c9792d99bd333c6783ce56150d1b11e3  
33bbcecd5c54d7  
Status: Downloaded newer image for ubuntu:latest
```

Figure 2: Pull an Ubuntu image from the Docker hub